

Protégé

Learn, Teach, Verify, Improve

THEME

Reinventing Education with AI

DOCUMENT

The idea, the problem it addresses, and how AI is used to solve it

DATE

October 2026

Riya passes every quiz in her intro programming course. But when a friend asks her to explain recursion, she can't. She never noticed the gap, her teacher never saw it, and the final exam will show it only when it's too late to fix.

PROTÉGÉ IN ONE SENTENCE

Students teach an AI, and that teaching produces verified evidence of what they understand, which then reaches the teacher as a plan for the next class.

AT A GLANCE

- **Learn** through examples drawn from what you love, such as space, drawing, football or music.
- **Teach Pip**, an AI companion who plays a slightly confused learner. Pip asks naive questions, restates ideas slightly wrong and plants common misconceptions to see whether the student catches them. A hidden **Assessor** reads every explanation against a rubric and updates the student's **concept map**, keeping the exact quote behind every score.
- **Improve and verify**. A red concept sends the student back to the prerequisite that's actually missing, then returns them to where they were. A short **Capstone Check** built from the student's own map then confirms what the map shows, using fresh questions with no hints. *We don't remove assessment; we remove the guesswork.*
- **Teachers** see what most of the class misunderstands, how many students are affected and the evidence behind it. They also get a **Next-Class Brief**: the top misconceptions, an example to show and a question to check understanding afterwards.

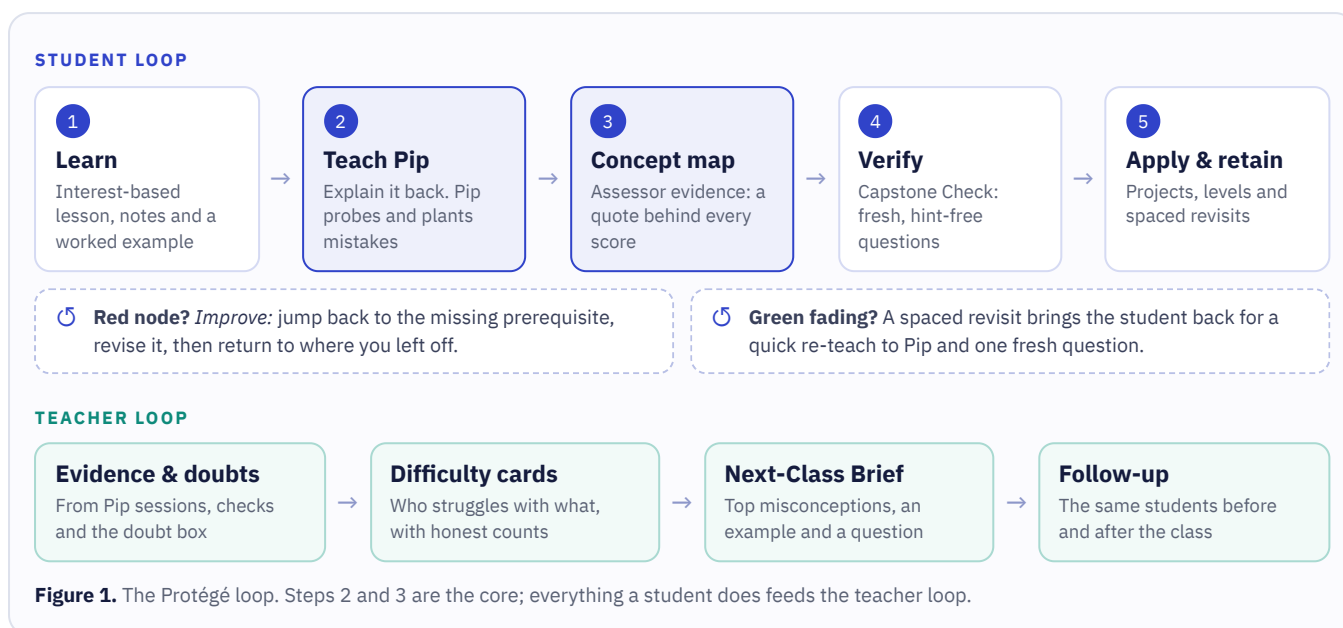
01 The problem

The theme asks what changes for students, for teachers and for how learning gets measured. Today all three fail at the same point: **a student's confusion rarely turns into a specific, timely next step.**

	What happens today	Why it matters
Students	Students in one class have different prerequisite gaps, interests and needs, yet they all get the same explanation at the same pace. When they struggle, the cause is often an earlier concept, and they may not know which one or may not want to ask.	Recognising an explanation feels like understanding it. Being able to repeat an answer is not the same as being able to explain, apply or remember it.
Teachers	They see marks, not reasoning. Nobody can investigate every difficulty in a class of 30 to 100+ students, or turn individual doubts into a plan for the whole class.	They reteach the wrong thing, or reteach the right thing to the wrong students.
Measurement	One big exam at the end gives one number, which says little about what the student misunderstands.	Gaps show up too late to fix, and nobody checks whether students still remember the material later.

AI adds a new risk. Students who get answers from unrestricted AI can do worse once the AI is taken away (Bastani et al., 2025). Adding a chatbot to the old classroom makes the problem worse.

02 The idea: one loop, four pillars



Pip and the evidence-backed concept map are the core. Everything else supports that loop:

Pillar	What's in it
Core loop	Learn → Teach Pip → Visible understanding → Improve → Verify
Personalisation	Interest profile, interest-based Learn mode, Explore projects
Levels	Pip's style and difficulty dial, revision levels from 1 up to Grandmaster
Retention	Spaced revisits (green fades to yellow over time) and streaks
Teacher insights	Student doubts sent to the dashboard, common difficulties, the Next-Class Brief and a follow-up check

03 Built from scratch, not bolted on

In a traditional classroom, learning and assessment are separate events, and feedback arrives weeks later. **In Protégé they are the same activity.** Every explanation a student gives Pip counts as evidence, and every piece of evidence leads straight to a next step. Pasting an answer won't get a student through, because Pip keeps asking follow-up questions. The big exam stays, but it becomes a **check on evidence already gathered rather than the final verdict.** The AI explains, converses and interprets. Teachers keep classroom judgement, and students stay active participants.

04 Meet Pip, and the Assessor behind Pip

Pip is an AI learning companion who plays a curious, slightly confused learner and wants the student to teach it. The idea comes from research on *teachable agents* and the *protégé effect*: students try harder to learn something when they have to teach it to someone else (Chase et al., 2009). Explaining something out loud exposes gaps that rereading hides.

How Pip behaves

- **One question per turn.** Pip asks naive questions, asks for examples and pushes back on incomplete reasoning: "Why does that call return first?" "What if the input is zero?"
- **Restates things slightly wrong.** Pip paraphrases with small errors, and the student has to notice and correct them.
- **Spot Pip's mistake.** Sometimes Pip confidently states a common misconception: "So when we hit the base case, that's the final answer and everything stops, right?" If the student agrees, that reveals the misconception. If the student corrects Pip, that shows they really understand.

- **Style and difficulty dial.** The student picks Pip's personality (**curious beginner**, **skeptical friend** or **tough interviewer**) and, separately, how deep Pip probes (Level 1 to 3). Tone changes how Pip talks, and the level changes what Pip asks. **Neither changes the grading standard.**
- **Hint ladder when the student is stuck.** Pip helps in four steps: **a nudge question → a pointer to the study notes → an analogy → a partial explanation, only as a last resort.** Each step used is logged and lowers the score for that explanation. Getting stuck counts as evidence in its own right.
- **Pip never does the student's work.** If a student asks Pip to just give the answer, Pip says "*Hey, I'm the one learning here!*" and points them to Learn mode.
- **Pip's notebook and session card.** Pip writes down what it has been taught (*base case ✓, recursive step ✓, moving toward the base case ✗*). At the end of a session, a short card lists what the student explained well, what tripped them up and one thing to try next.

The hidden Assessor. A separate AI agent **never talks to the student**. After each explanation, it scores the explanation against the concept's rubric and returns a score, a confidence level, **the exact quote that justifies the score**, any misconception it suspects and a likely missing prerequisite. Pip is tuned to be a natural conversation partner, while the Assessor is kept strict and consistent, so the friendly side never grades and the grader never has to be friendly.

A sample exchange (Riya picked *football*):

Riya: Recursion is like a knockout tournament. To find the champion, you find the winners of each half of the bracket.

Pip (*skeptical friend*): Okay, but when does it stop? Do we keep splitting forever?

Riya: It stops when a bracket has just one team. That team is its own winner.

Pip: Oh, so that's the base case! (*notebook: base case ✓*) What would happen if we forgot it?

What Pip means for teachers. Pip lets teachers see dozens of private explanations without reading any of them. Each one becomes structured evidence with a concept, a misconception and a supporting quote. The dashboard then shows which concepts students can't explain, and why.

05 The student journey

5.1 Sign-up

Students pick their interests (**space, drawing, football, music**) and an entry path. **"I'm new"** starts with a lesson, so a complete beginner is never stuck. **"Check what I know"** starts with a short check of what they already know.

5.2 Learn: the on-ramp

For each concept, the tutor gives a short explanation, one worked example and a quick check question, all based on reviewed course notes. One-tap buttons ask for **an analogy**, **a simpler version** or **code**. The tutor explains concepts, but it never does the student's work or answers its own check questions. Explanations use the student's interest as a lens:

Interest	Recursion explained as...
Space	A rocket's staged countdown: each stage fires, then hands off to a smaller stage until liftoff
Drawing	Nested frames or a fractal tree: each branch draws smaller branches until they're too small to draw
Football	A knockout bracket: the champion is the winner of two half-brackets, all the way down to a single team
Music	A cumulative song: each verse adds a line, then the song unwinds back through every earlier line, the same way function calls return

Each analogy ends with **"Where this analogy breaks"** (for example: "*In a real tournament, matches run at the same time. In recursion, each call waits for the one below it to finish.*"). If an analogy would mislead, the student gets a plain explanation instead.

5.3 Teach Pip, and watch the map update

The student explains the concept to Pip (Section 4). Their **personal concept map** updates live. Each node is **red** (a gap or misconception), **yellow** (shaky), **green, dashed outline** (explained well but not yet verified) or **solid green ✓** (verified by a check). Clicking "Why this score?" on any node shows the evidence quote and the rubric reason behind it.

5.4 Improve: jump back to the real gap

When a concept turns red, Protégé doesn't make the student redo the whole chapter, and it doesn't assume every earlier concept is weak. It follows the map's prerequisite links to the most likely gap (for example, *return values*) and gives a two-question check. If that's where the gap is, the student revises it and then **returns exactly where they left off**. If the prerequisite is fine, Protégé looks more closely at the current concept instead.

5.5 Verify: the Capstone Check (the big exam, redesigned)

When the student is ready, they take a short Capstone Check **built from their own map**. Weak concepts (red or yellow) get two questions from different question types, and green concepts get a spot-check. There are **no hints and no Pip**. Question types are short answer, "explain why" and "**find the error in this explanation**", which reuses Pip's planted misconceptions. Before submitting, the student taps how sure they are: *sure / think so / guessing*. A confident wrong answer points to a misconception rather than a simple gap, and correcting it tends to stick better (Butterfield & Metcalfe, 2001).

The check **adjusts the map rather than overwriting it**. The student sees the map before and after, and **places where the check and the map disagree are shown openly**. This keeps the system honest and lets us measure how reliable the Assessor is.

5.6 Predict, then run

For one recursion concept, the student predicts what `countdown(3)` prints before a visualiser runs it. The animation shows calls stacking up and returning, and **highlights the first step where the prediction was wrong**. Rules memorised without understanding don't survive this.

5.7 Apply and retain

- **Revision levels:** Level 1 Basic → Level 2 Application → Level 3 Challenge → **Grandmaster**. Grandmaster is a debugging or edge-case task defended against tough-interviewer Pip. Passing it earns a **Grandmaster badge, title and certificate**.
- **Explore projects:** when a concept is verified, the student unlocks a small project in their interest area. Examples include a **recursive drawing generator** (inspired by an artist's patterns), a **knockout-bracket builder**, a **rocket-countdown simulation** or a **recursive rhythm pattern**. Projects are curated, not generated on the fly, so the facts are right. Any physical project (such as model rockets) includes safety notes and a reminder to work with adult supervision.
- **Spaced revisits:** following the Ebbinghaus forgetting curve and spaced repetition, a verified green node **slowly fades toward yellow** (reviews at 1, 3, 7, 14 and 30 days). A quick re-teach to Pip plus one fresh question makes it green again.
- **Streaks** count days with real learning activity, not logins. They are shown as engagement, **never as proof of learning**.

06 What changes for teachers

- **Doubts reach the dashboard.** Students can ask the AI any doubt. The AI answers it and logs the doubt as a short structured note for the teacher (concept, difficulty, a supporting phrase). The interface tells students clearly that **doubts are shared with their teacher, without their name unless they choose to add it**. This way it doesn't feel like surveillance, and quiet students still get heard.
- **What most students struggle with.** Ranked cards show honest counts and **separate students who reported confusion from those whose assessed answers show the gap**:

"The base case is the final answer": 9 of 24 assessed students (6 were confident), 4 more reported confusion, 2 recovered after revision. [Open evidence](#) →

- **Class heatmap.** The same concept map, coloured by how much of the class has verified each concept.
- **Peer-teaching suggestions.** Students who have verified a concept are paired with classmates who are red on it, so students teach each other as well.
- **The Next-Class Brief.** One click produces a half-page plan:

Next class: Recursion (Section B, 24 of 31 assessed)

1. **"The base case is the final answer"** (9 students). *Show:* trace `factorial(3)` with a call stack of sticky notes and ask "Who does `factorial(1)` return to?"
2. **"No progress toward the base case"** (5 students). *Show:* a countdown that never ends, then fix it together.
3. **"Prints after the call run before it"** (4 students). *Show:* the cumulative-song example.

Check afterwards: "What does `mystery(3)` print?" (*from the reviewed question bank*)

- **Did it work?** After the class, the follow-up question compares **the same students before and after**. Students who haven't answered are listed separately, and the AI never claims the lesson *caused* any improvement.
- **The misconception list grows.** Doubts that don't match any known misconception are grouped and proposed to the teacher as new entries. The teacher approves each one before it's added.

07 How learning gets measured

Measurement is built around **what students can do on their own, how well they retain it, and whether teachers can act on the evidence.**

What we measure	How	What it tells us
Independent understanding	Matched questions before learning vs. fresh questions after, without hints	Whether understanding goes beyond the guided activity
Ability to explain	A consistent rubric covering correctness, reasoning, examples and handling follow-ups	Whether students can say <i>how</i> and <i>why</i> a concept works
Misconception recovery	Follow a flagged gap through revision to a <i>different</i> follow-up question	Whether that specific gap was addressed
Application and transfer	Use the concept in an unfamiliar problem or project	Whether understanding goes beyond memorised examples
Retention	Fresh questions after a delay (spaced revisits)	Whether learning lasts
Independence from help	Hint-ladder use alongside performance on comparable tasks	Whether students need less support over time
Assessor reliability	Agreement with hand-labelled dialogues, and how often the map and the Capstone Check agree	Whether the measurement itself can be trusted
Teacher usefulness	Teacher ratings of the identified gaps and briefs, plus prep-time feedback	Whether the insights are accurate and actionable
Classroom follow-up	The same students' answers before and after a targeted class activity	Whether the difficulty went down
Participation	Sessions, scheduled reviews, return visits	Whether students keep a routine (engagement, not proof of learning)

Longer-term learning gains would need a pilot with real students, comparable assessments and a delayed follow-up.

08 How AI is used, and the guardrails

AI agent	What it does	What it never does
Tutor	Explains concepts through the student's interest, gives worked examples, answers doubts, writes session cards	Answer its own check questions or do the student's work
Pip	Has the teaching conversation: probes, restates things slightly wrong, plants misconceptions, runs the hint ladder	Give correct answers directly or grade anything
Assessor (hidden)	Scores each explanation against a fixed rubric, with an evidence quote, a misconception flag and a likely missing prerequisite	Talk to the student
Check grader	Grades "explain why" answers in the Capstone Check against the rubric, with evidence	Grade objective items, which code scores
Teacher insights	Summarises gaps and doubts, drafts the Next-Class Brief and follow-up summary	Invent numbers, since every count is computed from the database

The concept graph (concepts, prerequisites, rubrics, misconceptions, notes) is drafted with AI and then **checked by hand** before use, and teachers can edit it. Mastery colours, review dates, streaks, unlocks and class statistics are computed by **code from saved evidence**. Every score shows its quote and rubric reason.

09 Risks and mitigations

Risk	Mitigation
AI scoring can be inconsistent	A fixed rubric, structured output, an evidence quote for every score, a low-temperature Assessor, and the Capstone Check as a cross-check
Students who haven't learned the topic can't teach it	Learn mode comes first, the hint ladder handles stuck moments, and getting stuck counts as evidence
Students paste in AI-written explanations	Pip's follow-up questions and planted-misconception traps are hard to pass without real understanding
The tutor or Pip slips into giving answers	Strict prompts, plus an output check that blocks full solutions to check and exam items. The model never sees answer keys for live checks.
Students feel monitored	A clear sharing notice, anonymous doubts by default, and students can see all their own evidence
Engagement features distort learning	Streaks and badges are shown as engagement and never feed into mastery scores
Physical project safety and factual accuracy	Only curated projects, with safety notes and adult-supervision reminders

10 What sets Protégé apart

- 1. The student teaches, and the AI learns.** Most AI tutors leave the student passive. Pip makes the student do the thinking, and planted mistakes test whether the understanding is real.
- 2. Teaching is the measurement.** Evidence comes from how students reason, every score links to a quote, and "Why this score?" is one click away.
- 3. The exam checks the system instead of only grading the student.** The Capstone Check is built from each student's own map, and openly shows where it disagrees with the map.
- 4. It fixes the root cause, not the symptom.** Protégé jumps back to the missing prerequisite and then returns the student to the original task, instead of making them reread the chapter.
- 5. Interests run through everything.** Lessons, Pip's examples, projects and reviews all use the student's interests, and every analogy says where it breaks.

6. **Teachers get an action, not a percentage.** Instead of "class mastery: 62%", they get "9 students believe X, here's the evidence, here's what to do tomorrow, and here's how to check it worked."
7. **Safe to ask.** Private doubts reach the teacher anonymously, so quiet students count too.

11 Hackathon scope (Round 2, Oct 10)

We will demonstrate **one complete journey** on a focused set of 8 introductory programming concepts. A student learns, teaches Pip, reveals a gap, revises a prerequisite and passes an independent check, while the teacher receives the evidence and a suggested classroom response.

	Features
Core (will ship)	Learn mode with notes and explanation styles, Teach Pip with the hint ladder, the Assessor, the live concept map with "Why this score?", the prerequisite jump, a simplified 5-question Capstone Check, and the teacher dashboard with the Next-Class Brief
Light add-ons (ship once the core is stable)	Interest profile and interest-based explanations, Pip's style and level dial, spaced decay, streaks
Stretch	2 or 3 curated Explore projects, higher revision levels and the Grandmaster badge, predict-then-run for one concept, peer-teaching suggestions
Future work	Voice teaching, a code sandbox, teacher-uploaded course material, regional-language support, a full interest-exploration space (artist profiles, step-by-step project guides), full gamification (XP, leaderboards), fully adaptive exams, a calibration dashboard, Bayesian Knowledge Tracing

12 Research basis

- **Bloom (1984)**, *The 2 Sigma Problem*, Educational Researcher 13(6). One-on-one tutoring produces large gains, and the challenge is doing it at scale.
- **Chase, Chin, Oppezzo & Schwartz (2009)**, *Teachable agents and the protégé effect*, J. Science Education & Technology 18(4). Biswas et al. (2005), *Betty's Brain*. The basis for Teach Pip.
- **Bastani et al. (2025)**, *Generative AI without guardrails can harm learning*, PNAS. Why the AI never does the student's work.
- **Kestin et al. (2025)**, a randomised study in which a carefully designed AI tutor outperformed in-class active learning in physics (Scientific Reports).
- **Corbett & Anderson (1994)**, *Knowledge tracing*; **Piech et al. (2015)**, *Deep knowledge tracing*. The basis for the mastery estimate.
- **Ebbinghaus (1885)** and **Dunlosky et al. (2013)**, *Effective learning techniques*. The basis for spaced revisits and practice testing.
- **Butterfield & Metcalfe (2001)**, *Errors committed with high confidence are hypercorrected*. The basis for the confidence tap.
- **Walkington (2013)**, *Personalising instruction to student interests*, J. Educational Psychology 105(4). The basis for interest-based explanations.

This research supports the design, but our specific implementation still needs to be tested with real students. The measurement plan in Section 7 is how we would do that.

Companion document: **Protégé • Technical Documentation**. It covers the architecture, the AI models and the Cline build plan.